

复流形上的几何 III

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1 Lecture 3

Torsion forms and Bott-Chern secondary class

Chern connection

omitted.

Torsion forms for Chern complex

Let $(E, v) : 0 \rightarrow E_1 \xrightarrow{v} \cdots \xrightarrow{v} E_m \rightarrow 0$ be a holomorphic chain complex of finite dimensional hol vector bundles on a connected complex manifold X , i.e. E_j are holomorphic vector bundles and $v : E_j \rightarrow E_{j+1}$ are $\mathbb{C}$ -linear holomorphic map. (If s is a local hol section of E_j on U , then $(vs)_x := v_x s_x$ is a local holomorphic section of E_{j+1} on $U \iff \bar{\partial}^E v + v \bar{\partial}^E = 0$). Chain complex: $v^2 = 0 \iff (v + \bar{\partial}^E)^2 = 0$ on $\Omega^{\cdot,\cdot}(X, E)$.

Set $E_+ = \bigoplus_{j \text{ even}} E_j$, $E_- = \bigoplus_{j \text{ odd}} E_j$, $E = E_+ \oplus E_-$. Let $h^E = \bigoplus_j h^{E_j}$ be a Hermitian metric on E . So $E_i \perp E_j$ if $i \neq j$.

Let v^* be the adjoint of v w.r.t. h^E . Set $V = v + v^*$. Then $\nabla^E + V$ is a superconnection on E .

Let

$$P^X := \bigoplus_p \Omega^{p,p}(X), \quad P^{X,0} := \{\omega \in P^X \mid \exists \eta, \eta' \in \Omega(X) \text{ s.t. } \omega = \partial\eta + \bar{\partial}\eta'\}$$

命题 1.1. Set

$$A = \bigoplus_{p,q,j} \Omega^{p,q}(X, \text{Hom}(E_j, E_{j+p-q})),$$

then A is a super-algebra. Moreover, if $\eta \in A$, then $\text{Tr}_s[\eta] \in P^X$.

证明. If $\eta, \eta' \in A$, then $\eta \hat{\otimes} \eta' \in A$. Now Tr_s vanishes on $\text{Hom}(E_j, E_k)_{j \neq k}$, thus $\text{Tr}_s[\eta] \in P^X$. $\square$

Let

$$A_u = \nabla^E + \sqrt{u}V, \quad u \geq 0.$$

be a superconnection on E . Let N be a number operator on E , $N|_{E_j} = j\text{Id}_{E_j}$.

定理 1.2. For any $u \geq 0$, $\text{Tr}_s[\exp(-A_u^2)]$, $\text{Tr}_s[N \exp(-A_u^2)]$ are in P^X . The form $\text{Tr}_s[\exp(-A_u^2)]$ is closed. For $u > 0$,

$$\frac{\partial}{\partial u} \text{Tr}_s[\exp(-A_u^2)] = -\frac{1}{2u}(\partial + \bar{\partial})\text{Tr}_s[\sqrt{u}V \exp(-A_u^2)],$$

$$\mathrm{Tr}_s[\sqrt{u}V \exp(-A_u^2)] = (\partial - \bar{\partial})\mathrm{Tr}_s[N \exp(-A_u^2)].$$

In particular,

$$\frac{\partial}{\partial u}\mathrm{Tr}_s[\exp(-A_u^2)] = -\frac{1}{u}\bar{\partial}\partial\mathrm{Tr}_s[N \exp(-A_u^2)].$$

证明. Step 1: Show that $A_u^2 \in A$.

Since

$$A_u^2 = (\nabla^E)^2 + uV^2 + \sqrt{u}[\nabla^E, V],$$

where

$$\begin{aligned} (\nabla^E)^2 &\in \Omega^{1,1}(X, \mathrm{End}(E)), \\ V^2 &= (v + v^*)^2 = vv^* + v^*v \in C^\infty(X, \mathrm{End}(E)), \end{aligned}$$

and

$$[\nabla^E, V] = [\nabla'', v] + [\nabla', v] + [\nabla'', v^*] + [\nabla', v^*]$$

By the definition of holomorphic chain complex,

$$[\nabla'', v] = \bar{\partial}v + v\bar{\partial} = 0.$$

For $u \in T^{1,0}X$,

$$\begin{aligned} \langle ([\nabla'_u, v^*]\sigma_1, \sigma_2) \rangle &= \langle (\nabla'_u v^* - v^* \nabla'_u)\sigma_1, \sigma_2 \rangle \\ &= \langle \nabla'_u(v^* \sigma_1), \sigma_2 \rangle - \langle \nabla'_u \sigma_1, v \sigma_2 \rangle \\ &= u \langle v^* \sigma_1, \sigma_2 \rangle - \langle v^* \sigma_1, \nabla''_{\bar{u}} \sigma_2 \rangle - u \langle \sigma_1, v \sigma_2 \rangle + \langle \sigma_1, \nabla''_{\bar{u}}(v \sigma_2) \rangle \\ &= \langle \sigma_1, (\nabla''_{\bar{u}} v - v \nabla''_{\bar{u}})\sigma_2 \rangle \\ &\stackrel{?}{=} 0. \end{aligned}$$

So

$$[\nabla^E, V] = [\nabla', v] + [\nabla'', v^*]$$

where

$$[\nabla', v] \in \Omega^{1,0}(X, \mathrm{End}(E_j, E_{j+1})), \quad [\nabla'', v^*] \in \Omega^{0,1}(X, \mathrm{End}(E_j, E_{j-1}))$$

Thus $A_u^2 \in A$, which deduces $\exp(-A_u^2) \in A$. With $N \in A$, we conclude that $\mathrm{Tr}_s[\exp(-A_u^2)], \mathrm{Tr}_s[N \exp(-A_u^2)] \in P^X$.

Step 2: Verify the formulas via Duhamel's formula.

引理 1.3. For $H_u \in (\wedge(T^*X) \otimes \text{End}(E))^+$

$$\frac{\partial}{\partial u} \exp(-tH_u) = - \int_0^t e^{-(t-s)H_u} \left(\frac{\partial}{\partial u} H_u \right) e^{-sH_u} ds.$$

Proof of lemma. omitted. □

Now

$$\begin{aligned} \frac{\partial}{\partial u} \text{Tr}_s [\exp(-A_u^2)] &= -\text{Tr}_s \left[\int_0^1 e^{-(1-s)A_u^2} \left(\frac{\partial}{\partial u} A_u^2 \right) e^{-sA_u^2} ds \right] \\ &= - \int_0^1 \text{Tr}_s \left[\left(\frac{\partial}{\partial u} A_u^2 \right) e^{-(1-s)A_u^2} e^{-sA_u^2} \right] ds \quad \text{Tr}_s[A, B] = 0 \\ &= - \text{Tr}_s \left[\frac{\partial}{\partial u} A_u^2, e^{-A_u^2} \right] \\ &= - \text{Tr}_s \left[\left[A_u, \frac{1}{2\sqrt{u}} V \right], e^{-A_u^2} \right] \quad \frac{\partial}{\partial u} A_u^2 = \left[A_u, \frac{\partial}{\partial u} A_u \right], \frac{\partial}{\partial u} A_u = \frac{1}{2\sqrt{u}} V \\ &= - \text{Tr}_s \left[A_u, \frac{1}{2\sqrt{u}} V e^{-A_u^2} \right] \quad [A_u, A_u^2] = 0 \\ &= - d\text{Tr}_s \left[\frac{1}{2\sqrt{u}} V e^{-A_u^2} \right] \end{aligned}$$

□